DATA ANALYTICS PORTFOLIO

CRIME INVESTIGATION ANALYTICS

Los Angeles | 2020–2024

Power BI · Advanced DAX · 800K+ Records

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OBJECTIVES

Project Objectives

An analytical platform built to investigate crime patterns and support decisions, not simply a dashboard.

01

Map Crime Hotspots

Surface high-concentration crime zones across all 21+ LAPD divisions.

02

Analyze Trends

Track year-over-year and seasonal shifts across five years of data.

03

Surface Patterns

Break down offense types, weapons, and victim demographics at a glance.

04

Drive Decisions

Turn raw incident data into evidence-based actions for resource and policy decisions.

THE PROBLEM

How do you find meaningful connections inside more than 800,000 crime records?

- Thousands of incidents across 21+ divisions
- Overlapping offenses and case relationships
- Patterns hidden inside five years of data



CRIME RECORDS

THE APPROACH

Raw data becomes investigation-ready intelligence.

01

Raw Crime Data

Power Query · cleaning

→

02

Data Model

Snowflake schema · DAX

→

03

Investigation Analytics

Geographic · temporal · relational

→

04

Actionable Insights

Evidence-based decisions

DATA PROFILE

Dataset Overview

CRIME RECORDS

800K+

Incident-level data sourced directly from the LA Open Data Portal.

2020–2024

ANALYSIS PERIOD

Five full years of reported crime across Los Angeles.

21+

LAPD DIVISIONS

Complete geographic coverage of all LAPD reporting areas citywide.

30+

CRIME CATEGORIES

From violent offenses to property and quality-of-life crimes.

15+

ATTRIBUTES / RECORD

Date, location, weapon, victim demographics, and case status per incident.

Source: LA Open Data Portal

Publicly available and regularly updated.

INVESTIGATION ANALYTICS

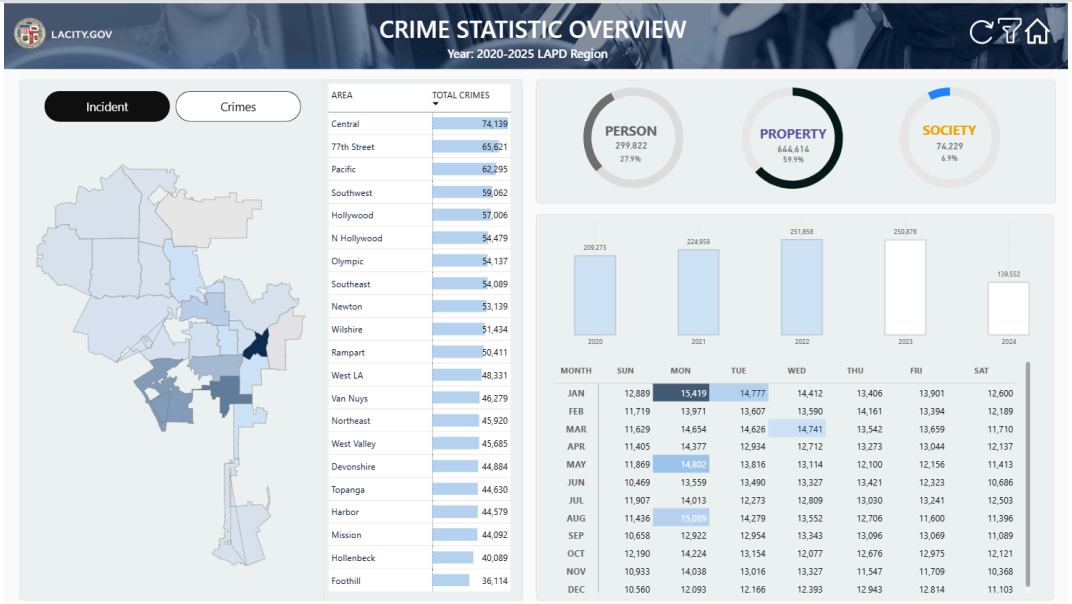
Investigators move from a single case to related cases in seconds.

01	**Search Case**	Surface a case's full detail record by case number.
02	**Identify Offense Pattern**	Break down the offenses tied to the selected case.
03	**Find Related Cases**	A configurable proximity radius surfaces nearby incidents.
04	**Analyze Location & Time**	Multi-level view, from city-wide down to individual coordinates.

EXECUTIVE DASHBOARD

Executive Dashboard: Page 1 Overview

Power BI



Interactive LAPD Map

Surfaces geographic crime hotspots across divisions in real time.

KPI Cards

Crime volume broken down by persons, property, and society categories.

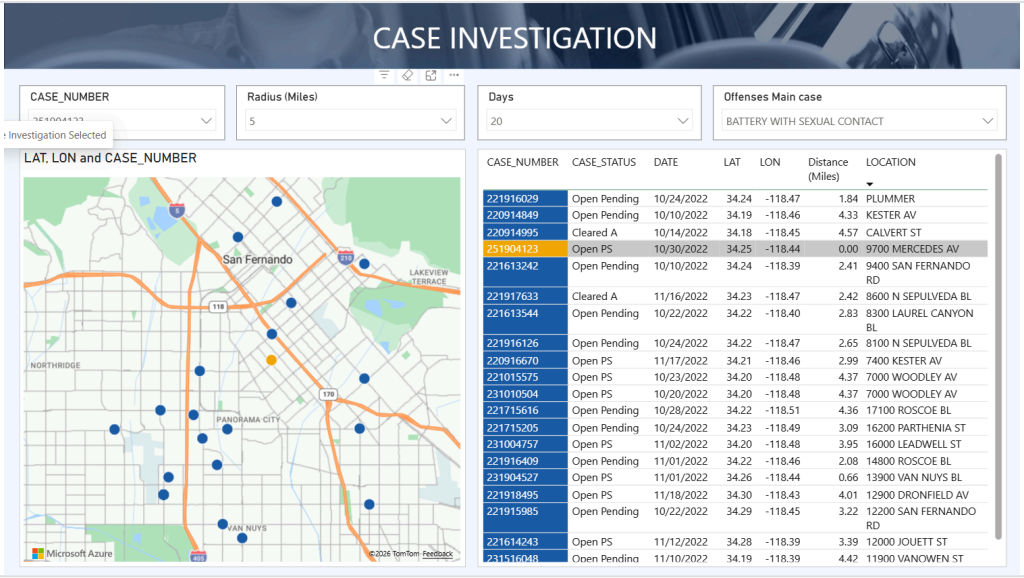
Trend & Heatmap

Year-over-year chart and monthly heatmap reveal seasonal patterns from 2020–2024.

CASE INVESTIGATION

Interactive Case Investigation

Power BI



Case Number Search

Instantly surfaces a specific incident's full detail record by case number.

Proximity Analysis

Configurable radius buffer maps all nearby incidents and populates a filtered table.

Multi-Level Analysis

From city-wide executive summaries down to individual case coordinates and active investigation.

FINDINGS

Key Insights

PEAK YEAR

Crime peaked in 2022

GEOGRAPHY

Central had the highest crime volume

TIMING

Monday showed high activity across several months

CATEGORY

Property crime was the largest category

These insights can help decision-makers understand where and when crime is more concentrated and support better resource allocation.

FROM ANALYTICS TO ACTION

Business Recommendations

Evidence-based actions derived directly from the data.

01 **Focus Resources on High-Volume Areas**

Use crime-volume patterns to prioritize areas requiring greater attention and resources.

02 **Target High-Activity Periods**

Use temporal patterns to better align patrol and operational resources with periods of elevated activity.

03 **Prioritize Property Crime**

Because property crime represents the largest category, targeted strategies can help address the largest area of demand.

04 **Monitor Trends Over Time**

Continue tracking crime trends to evaluate whether changes are sustained and to support future resource planning.

CAPABILITIES

Skills Demonstrated

Extraction → Model → Analysis → Story

SQL

Data extraction, filtering, and aggregation from relational databases.

Power Query

End-to-end data cleaning, transformation, and loading into the BI model.

Data Modeling

Snowflake schema design with optimized fact and dimension table relationships.

DAX

KPIs, year-over-year measures, and dynamic context-aware filtering built from scratch.

Power BI

Multi-page interactive dashboard with drill-through, cross-filtering, and live KPI cards.

Data Storytelling

Translating 800K+ records into insights executives can understand and act on.

THANK YOU

From raw crime records to decisions that matter.

Crime Investigation Analytics

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